

Requirements

Cohort 1 Group 7

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Introduction

The requirements were gathered from the product brief and a customer meeting. First, the product brief was read and core deliverables were collated. A meeting was then organised with the customer where the deliverables were discussed and refined with more understanding. The customer also added additional requirements that were not present in the product brief and a single statement of need was gathered. After the meeting the user requirements were imputed into a table with an unique ID and how important the feature is; a table was used because it makes it clear and concise for anyone involved in the project to understand and use the data. From the user requirements, another two tables were created for system requirements: functional and non-functional, each with the ID from the user requirements table, another unique ID and a description that can be measured. These are also presented in tables as it allows the developers to code and test each requirement quickly and easily ensuring all essential deliverables are met.

Statement of user and system requirements

SSON - A single-player desktop game that doesn't last more than 5 minutes and has the player escaping a situation at university

User requirements table

<u>ID</u>	<u>Description</u>	<u>Priority</u>
UR_SINGLEPLAYER	The game shall be singleplayer	Shall
UR_TIME	The game shall not take longer than 5 minutes	Shall
UR_DEAN	The game shall have a dean who chases the player	Shall
UR_NEGATIVE_EVENTS	The game shall have 5 negative visible events that disadvantage the player	Shall
UR_POSITIVE_EVENTS	The game shall have 5 positive visible events that advantage the player	Shall
UR_HIDDEN_EVENTS	The game shall have 3 hidden events that either advantage or disadvantage the player	Shall
UR_ESCAPE	The game shall have a way for the player to escape	Shall
UR_PAUSE	The game shall have a way to pause the game	Shall
UR_SCORING_SYSTEM	The game shall have a scoring system based on time and events	Shall
UR_OPERATING_SYSTEM	The game shall be playable on all operating systems	Shall
UR_TUTORIAL	The game should have an introductory scene to give the player a tutorial	Should
UP_MAP	The game should consist of a single large map	Should
UR_STYLE	The game should be played from a top down perspective	Should
UR_POWER_UP	The game should give the player abilities from events	Should
UR_STAMINA	The game may have a stamina bar that limits the players speed	May
UR_DIFFICULTIES	The game may have different difficulties	May

Functional requirements table

<u>ID</u>	<u>Description</u>	<u>User Requirements</u>
FR_SINGLEPLAYER	The game will only support singleplayer	UR_SINGLEPLAYER
FR_FIVE_MINUTE	The game will not allow the player to play for more than 5 minutes	UR_TIME
FR_EXIT_FIVE	The game may end before 5 minutes if the players finds an exit	UR_TIME
FR_DEAN_CHASE	The game will have a dean that chases the player	UR_DEAN
FR_DEAN_CATCH	The dean will not end the game	UR_DEAN
FR_DEAN_LIMIT	The dean will limit the players movement	UR_DEAN
FR_NEGATIVE	The game will show the player at least 5 choices that disadvantage them	UR_NEGATIVE_EVENTS
FR_POSITIVE	The game will show the player at least 5 choices that advantage them	UR_POSITIVE_EVENTS
FR_HIDDEN	The game will have at least 3 events that have a hidden trigger	UR_HIDDEN_EVENTS
FR_ESCAPE	The game has a way for the player to win by escaping the map	UR_ESCAPE
FR_PAUSE	The game will include a pause function to stop gameplay and timer	UR_PAUSE
FR_SCORING_SYSTEM	The game will calculate a score based on elapsed time and triggered events	UR_SCORING_SYSTEM
FR_OS_SUPPORT	The game will be playable across major operating systems	UR_OPERATING_SYSTEM
FR_TUTORIAL_SCENE	The game should include an introductory tutorial scene to teach controls and objectives	UR_TUTORIAL
FR_SINGLE_MAP	The game should consist of one large continuous map	UP_MAP
FR_TOP_DOWN_VIEW	The game should use a top-down camera perspective	UR_STYLE
FR_POWER_UPS	The game will provide the player with special abilities through events	UR_POWER_UP
FR_STAMINA_BAR	The player may have a stamina bar that limits movement speed	UR_STAMINA
FR_DIFFICULTY_MODES	The game may include multiple difficulty modes	UR_DIFFICULTIES

Non functional requirements table

<u>ID</u>	<u>Description</u>	<u>User Requirements</u>	<u>Fit Criteria</u>
NFR_TIME_LIMIT	The game must ensure that a complete play session does not exceed 5 minutes	UR_TIME	Each game session automatically ends after 5 minutes of gameplay
NFR_HIDDEN_EVENTS	The game must display a hidden event when it is triggered	UR_HIDDEN_EVENTS	The hidden events will show within 1 second of the trigger
NFR_OS_COMPATIBILITY	The game shall run on all major operating systems (Windows, macOS, Linux)	UR_OPERATING_SYSTEM	The game launches and runs without errors on all three OS environments during testing
NFR_OS_PERFORMANCE	The game shall maintain consistent performance and frame rate across all supported operating systems	UR_OPERATING_SYSTEM	Average frame rate $\geq$ 30 FPS on standard hardware across all tested OS versions
NFR_VISUAL_STYLE	The game should use a top-down perspective to maintain visual clarity and consistency	UR_STYLE	Gameplay camera remains fixed above the player perpendicular to the ground
NFR_USER_INTERFACE	The user interface should be intuitive and minimalistic to support top-down gameplay	UR_STYLE	90% of test users can navigate the UI and start a game without external guidance
NFR_STABILITY	The game shall not crash or freeze during normal operation	UR_OPERATING_SYSTEM	0 critical crashes over 10 consecutive games
NFR_LOAD_TIME	The game should load completely within 5 seconds on a standard desktop computer	UR_OPERATING_SYSTEM	Average load time $\leq$ 5 seconds measured over 5 startup tests on standard hardware
NFR_USABILITY_TUTORIAL	The tutorial should be clear and concise, enabling a new player to learn controls quickly	UR_TUTORIAL	80% of first-time players complete the tutorial and start gameplay within 2 minutes